

Advanced Statistical Mechanics

Assignment 01

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Question 1 (Yang-Lee edge):

Consider the following grand partition function for a hypothetical fluid

$$\Xi(z, V) = (1 + z)^{[\alpha V]} (1 + z^{[\alpha V]}).$$

Here $[x]$ denotes the integer part of x .

- (a) Determine the zeros of $\Xi(z, V)$ and (roughly) plot them on the complex z plane. Here z denotes the fugacity, $z = e^{\beta\mu}$.
- (b) Show that the zeros of $\Xi(z, V)$ in the thermodynamic limit hit the real z axis at $z = 1$.
- (c) Evaluate

$$\Theta = \lim_{V \rightarrow \infty} \frac{1}{V} \ln \Xi(z, V)$$

for $z < 1$ and $z > 1$ and show that it is continuous but non-analytic.

- (d) Given that the particle density is $n = z \frac{\partial \Theta}{\partial z}$, show that there is a density jump at the phase transition.
- (e) Using the fact $\beta p = \Theta$, derive the equations of state $p(n)$, for $z < 1$ and $z > 1$ by eliminating z from the expressions of $p(z)$ and $n(z)$.

- (a) To find the roots of Ξ , we set

$$(z + 1)^{[\alpha V]} = 0 \quad z^{[\alpha V]} = -1 \tag{1}$$

The real root is $z = -1$ with multiplicity $[\alpha V]$, which is physically irrelevant since the fugacity is $z = e^{\beta\mu} > 0$. The other complex roots come from $z^{[\alpha V]} = -1$, which are the roots of -1 , lying on the unit circle

$$z_m = e^{i\theta_m} \quad \theta_m = \frac{(2m+1)\pi}{[\alpha V]}, \quad m \in \{0, 1, \dots, [\alpha V] - 1\}$$

- (b) We see that the number of roots increases as we increase V . Consider the root with $m = 0$ and $n = [\alpha V] - 1$:

$$z_0 = e^{i\pi/[\alpha V]} \quad z_1 = e^{i\pi(2[\alpha V]-1)/[\alpha V]}$$

- (c) From the definition,

$$\begin{aligned} \Theta &= \lim_{V \rightarrow \infty} \frac{1}{V} \ln \Xi(z, V) \quad z \in \mathbb{R}_{\geq 0} \\ &= \lim_{V \rightarrow \infty} \frac{1}{V} \left[[\alpha V] \ln(1 + z) + \ln(1 + z^{[\alpha V]}) \right] \end{aligned} \tag{2}$$

Question 2 (Critical exponents in the mean field Ising model.):

Familiarize yourself with the definition of the critical exponents β , γ , and δ . Show that the specific heat for the mean field Ising model satisfies

$$\begin{cases} c = \frac{3Nk_B}{2}, & T < T_c, \\ c = 0, & T > T_c. \end{cases}$$

If the specific heat critical exponent α is defined via the relation

$$c \sim \left| \frac{T - T_c}{T_c} \right|^{-\alpha},$$

what value of α do we infer from the above analysis?

Answer. The partition function of the mean-field Ising model can be written as,

$$\mathcal{Z} = [2 \cosh(\beta J \gamma m + \beta h)]^N e^{-\beta \varepsilon_0} \quad \varepsilon_0 = \frac{JN\gamma m^2}{2}$$

Now, let us find the internal energy $U = -\partial_\beta \ln \mathcal{Z}$ for the zero-field case,

$$\begin{aligned} U &= -\frac{\partial}{\partial \beta} \left(-\frac{\beta JN\gamma m^2}{2} + N \ln 2 + N \ln \cosh(\beta J \gamma m) \right) \\ &= N \left[\frac{J\gamma m^2}{2} + \beta J \gamma m \frac{\partial m}{\partial \beta} - \tanh(\beta J \gamma m) \left(J \gamma m + \beta J \gamma \frac{\partial m}{\partial \beta} \right) \right] \end{aligned} \quad (3)$$

Using the self-consistent equation $m = \tanh(\beta J \gamma m)$ we get,

$$\begin{aligned} U &= N \left[\frac{J\gamma m^2}{2} + \cancel{\beta J \gamma m \frac{\partial m}{\partial \beta}} - J \gamma m^2 - \cancel{\beta J \gamma m \frac{\partial m}{\partial \beta}} \right] \\ &= -\frac{JN\gamma m^2}{2} \end{aligned} \quad (4)$$

The specific heat is defined as $c = \frac{\partial U}{\partial T}$ which can be written as,

$$c = \frac{\partial U}{\partial T} = \frac{\partial U}{\partial \beta} \times \frac{\partial \beta}{\partial T} = -k_B \beta^2 \frac{\partial U}{\partial \beta} = k_B \beta^2 \frac{JN\gamma}{2} \times 2m \times \frac{\partial m}{\partial \beta} \quad (5)$$

Now, consider the reduced emperature defined as $t = \frac{T_c - T}{T_c}$ where $k_B T_c = J\gamma$,

$$\frac{1}{1-t} = \frac{T_c}{T} = \frac{J\gamma}{k_B T} = \beta J \gamma \quad (6)$$

Hence the self-consistent equation becomes $m = \tanh\left(\frac{m}{1-t}\right)$ which can be expanded as,

$$m = \frac{m}{1-t} - \frac{m^3}{3(1-t)^3} + \dots \quad (7)$$

from which we get either $m = 0$ or $m^2 = (3t)(1-t)^2$. For $T > T_c$, we have only one solution $m = 0$ and hence $\frac{\partial m}{\partial \beta} = 0$ which implies that

$$c = 0 \quad \text{for } T > T_c \quad (8)$$

For $T < T_c$ we have,

$$m^2 = 3t + \mathcal{O}(t^2) \implies 2m \times \frac{\partial m}{\partial \beta} = 3 \frac{\partial t}{\partial \beta} = 3 \frac{\partial}{\partial \beta} \left(1 - \frac{1}{\beta k_B T_c} \right) = \frac{3}{k_B T_c \beta^2}$$

Substituting this in Eq. (4) we get for $T < T_c$,

$$c = \cancel{k_B \beta^2} \frac{JN\gamma}{2} \times \frac{3}{\cancel{k_B \beta^2} T_c} = \frac{3N}{2} \times \frac{J\gamma}{T_c} = \frac{3Nk_B}{2} \quad (9)$$

Thus, we obtain

$$c = \begin{cases} \frac{3Nk_B}{2} & T < T_c \\ 0 & T > T_c \end{cases} \quad (10)$$

Hence there is a discontinuous jump in specific heat across T_c . If we define the scaling

$$c \sim \left| \frac{T_c - T}{T_c} \right|^{-\alpha}$$

Then from the definition of the critical exponent, we have

$$\alpha = - \lim_{t \rightarrow 0^+} \frac{\ln(3Nk_B/2)}{\ln t} \rightarrow 0 \quad (11)$$

Hence, from the mean-field analysis, we obtain the critical exponent $\alpha = 0$.

Question 3 (q-state Potts model):

A generalization of the Ising model is the q-state Potts model, where one assume that the spin variable at a site can now take q different values, $\sigma_i \in \{1, 2, 3, \dots, q\}$ and the Hamiltonian is given by,

$$\mathcal{H}_{\text{Potts}} = -J \sum_{\langle ij \rangle} \delta_{\sigma_i \sigma_j} - h \sum_i \delta_{\sigma_i, 1}$$

where $\langle ij \rangle$ denotes nearest neighbour interaction. By defining a suitable transformation between the Ising spin variable σ_i and the Kronecker delta $\delta_{\sigma_i \sigma_j}$, show that the for $q = 2$, this Hamiltonian is equivalent to the Ising Hamiltonian.

Answer. The interaction term given here is $\delta_{\sigma_i \sigma_j}$ while in the Ising Hamiltonian, the interaction term is $S_i S_j$. If both Ising spins are different, the interaction is -1 while for the $q = 2$ Potts model, it is 0. If both Ising spins are same, the interaction is $+1$ which is same as in the $q = 2$ Potts model. Thus we can have a linearly transformation, such that

$$\delta_{\sigma_i \sigma_j} = a S_i S_j + b$$

The conditions to satisfy are,

$$0 = b - a \quad 1 = b + a$$

from which we obtain $a = b = 1/2$. Thus, the transformation is,

$$\delta_{\sigma_i \sigma_j} = \frac{1}{2}(S_i S_j + 1)$$

Essentially we are mapping $S_i = +1 \mapsto \sigma_i = 1$ and $S_i = -1 \mapsto \sigma_i = 2$. Thus the second Kronecker delta becomes,

$$\delta_{\sigma_i, 1} = \frac{1}{2}(S_i + 1)$$

With this transformation, the Potts Hamiltonian becomes,

$$\begin{aligned} \mathcal{H}_{\text{Potts}} &= -\frac{J}{2} \sum_{\langle ij \rangle} (S_i S_j + 1) - \frac{h}{2} \sum_i (S_i + 1) \\ &= -\frac{J}{2} \sum_{\langle ij \rangle} S_i S_j - \frac{h}{2} \sum_i S_i - \underbrace{\frac{J}{2} \sum_{\langle ij \rangle} 1 - \frac{h}{2} \sum_i 1}_{\text{const.}} \\ &= -J' \sum_{\langle ij \rangle} S_i S_j - h' \sum_i S_i + \text{const.} \\ &= \mathcal{H}_{\text{Ising}} + \text{const.} \end{aligned} \quad (12)$$

where $J' \equiv J/2$ and $h' \equiv h/2$. Since a constant in the Hamiltonian can be neglected (will cause only a shift in the energy), we obtain that for $q = 2$, the Potts model reduces to the Ising Hamiltonian.

Question 4 (Antiferromagnetic Ising Model):

Consider the Ising model on a two-dimensional square lattice with nearest neighbour interaction but with the interaction term having opposite sign such that antiparallel alignment is favoured.

$$\mathcal{H} = J \sum_{\langle ij \rangle} \sigma_i \sigma_j - h \sum_i \sigma_i \quad J > 0$$

- (a) Show that for $h = 0$, the partition function and hence the free energy is the same as that for the usual Ising model.
- (b) Solve the problem in the mean field approximation with $h = 0$.

Answer. (a) Let us divide the 2D lattice into two sub-lattices with sites A and site B , alternately placed. Hence, there are only sites of type B surrounding any site of type A (and vice versa). Let us consider the zero-field Ising model on this sublattice.

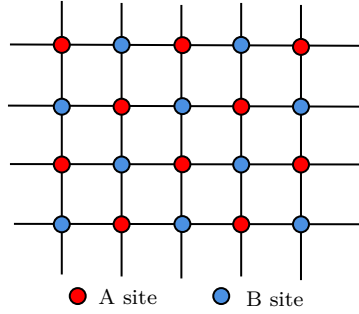


Figure 1: 2D lattice broken into interpenetrating sub-lattices.

Since interaction of spins on type A site is only with those at type B site, the Ising Hamiltonian $\mathcal{H}(h = 0, J, \{\sigma_i\}) = -J \sum_{\langle ij \rangle} \sigma_i \sigma_j$ can be written as,

$$\mathcal{H}(h = 0, J, \{\sigma_i^A\}, \{\sigma_i^B\}) = -J \sum_{\langle ij \rangle} \sigma_i^A \sigma_j^B \quad (13)$$

From this, the obvious symmetry exhibited by the Hamiltonian is,

$$\mathcal{H}(h = 0, -J, \{\sigma_i^A\}, \{\sigma_i^B\}) = \mathcal{H}(h = 0, J, \{-\sigma_i^A\}, \{\sigma_i^B\}) = \mathcal{H}(h = 0, J, \{\sigma_i^A\}, \{-\sigma_i^B\}) \quad (14)$$

The zero-field partition function becomes,

$$\begin{aligned} \mathcal{Z}(h = 0, -J, T) &= \sum_{\{\sigma^A\}} \sum_{\{\sigma^B\}} e^{-\beta \mathcal{H}(0, -J, \{\sigma^A\}, \{\sigma^B\})} \\ &= \sum_{\{\sigma^A\}} \sum_{\{\sigma^B\}} e^{-\beta \mathcal{H}(0, J, \{-\sigma^A\}, \{\sigma^B\})} \\ &= \sum_{\{\sigma^A\}} \sum_{\{\sigma^B\}} e^{-\beta \mathcal{H}(0, J, \{\sigma^A\}, \{\sigma^B\})} \\ &= \mathcal{Z}(h = 0, J, T) \end{aligned} \quad (15)$$

Going from the first to the second line, we used the sub-lattice symmetry from Eq. (13) while going from the second to the third line, we used the fact the since we are summing over all values of $\sigma_i = \pm 1$,

the sum with $\{-\sigma\}$ will be the same as the sum with $\{+\sigma\}$. Since the free energy is just the derivative of the partition function, we obtain

$$F(h = 0, J, T) = F(h = 0, -J, T) \quad (16)$$

Hence the partition function and the free energy for the ferromagnetic ($J > 0$) and the anti-ferromagnetic ($J < 0$) Ising model is same and hence, both these are thermodynamically equivalent.

(b) We apply the mean-field approximation where we neglect the fluctuations in the spin.

$$(m_A - \sigma_i^A)(m_B - \sigma_j^B) = m_A m_B - m_A \sigma_j^B - m_B \sigma_i^A + \sigma_i^A \sigma_j^B = 0 \quad (17)$$

Substituting this in the anti-ferromagnetic Hamiltonian with $J > 0$ we get,

$$\begin{aligned} \mathcal{H} &= J \sum_{\langle ij \rangle} \sigma_i^A \sigma_j^B = J m_A \sum_{\langle ij \rangle} \sigma_j^B + J m_B \sum_{\langle ij \rangle} \sigma_i^A - J m_A m_B \sum_{\langle ij \rangle} 1 - h \sum_i \sigma_i^A - h \sum_j \sigma_j^B \\ &= -(h - J m_A \gamma) \sum_i \sigma_i^B - (h - J m_B \gamma) \sum_i \sigma_i^A - \frac{J m_A m_B \gamma}{2} \\ &= -h_A \sum_i \sigma_i^B - h_B \sum_i \sigma_i^A - \frac{J m_A m_B \gamma}{2} \end{aligned}$$

The partition function becomes,

$$\mathcal{Z} = \mathcal{Z}_A^{N/2} \mathcal{Z}_B^{N/2} e^{-\beta J \gamma m_A m_B / 2}$$

where $\mathcal{Z}_A = 2 \cosh(\beta h_B)$ and $\mathcal{Z}_B = 2 \cosh(\beta h_A)$. The sublattice magnetisation can be obtained from the partition function as,

$$m_A = \tanh(\beta h_B) \quad m_B = \tanh(\beta h_A) \quad (18)$$

which are two coupled transcendental equations. In the zero field, $h_A = -J m_A \gamma$ and $h_B = -J m_B \gamma$. Expanding $\tanh(x)$ for $x \ll 1$ we get,

$$\begin{aligned} m_A &= -\tanh\left(m_B \frac{T_c}{T}\right) \simeq -\frac{T_c}{T} m_B + \frac{1}{3} \left(\frac{T_c}{T}\right)^3 m_B^3 \\ m_B &= -\tanh\left(m_A \frac{T_c}{T}\right) \simeq -\frac{T_c}{T} m_A + \frac{1}{3} \left(\frac{T_c}{T}\right)^3 m_A^3 \end{aligned} \quad (19)$$

Define $M := \frac{m_A + m_B}{2}$ and $m_s := \frac{m_A - m_B}{2}$ which allows us to write,

$$\begin{aligned} m_A^3 + m_B^3 &= 2M(M^2 + 3m_s^2) \\ m_A^3 - m_B^3 &= 2m_s(3M^2 + m_s^2) \end{aligned} \quad (20)$$

From Eq. (18), by adding and subtracting we have:

$$\begin{aligned} M \left(1 + \frac{T_c}{T}\right) &= \frac{M}{3} \left(\frac{T_c}{T}\right)^3 [M^2 + 3m_s^2] \\ m_s \left(1 - \frac{T_c}{T}\right) &= -\frac{m_s}{3} \left(\frac{T_c}{T}\right)^3 [3M^2 + m_s^2] \end{aligned} \quad (21)$$

The solution of the second equation for is $m_s = 0$ or h